

波函数叠加
一、简答题

(1) $\psi_2 = \psi_1 e^{i\phi}$

是波函数叠加的波函数，
叠加原理。

ψ_2 不一定是波函数的叠加原理的波函数，
因为如果 ψ_2 是波函数，那么叠加原理就不成立，
不是波函数的波函数，但叠加原理成立，
叠加原理的波函数。

(2)

$\psi = \psi_1 + \psi_2$

$P_1 = |\psi_1|^2$
 $P_2 = |\psi_2|^2$
 $P = |\psi|^2$

设

设 $\hat{L}\psi_1 = L_1\psi_1$

$\hat{L}\psi_2 = L_2\psi_2$

则波函数叠加原理的波函数
分别为：

$P_1 = |\psi_1|^2$
 $= |C_1|^2$

$P_2 = |\psi_2|^2$
 $= |C_2|^2$

波函数的叠加原理的波函数

(3) $\psi(r, t) = \sum_{n=1}^{\infty} C_n e^{-iE_n t/\hbar} \psi_n(r)$

二、解

$\psi = A(r + y + 2z) e^{-i\omega t}$
 $1107 = \frac{1}{\sqrt{2}} \cos \theta$
 $1117 = -\frac{1}{\sqrt{2}} \sin \theta$
 $11-17 = \frac{1}{\sqrt{2}} \sin \theta$

$\psi = A(r \sin \theta \cos \phi + r \sin \theta \sin \phi + 2r \cos \theta) e^{-i\omega t}$
 $= Ar \left[\frac{\sin \theta}{2} e^{i\phi} + \frac{1}{2} \sin \theta e^{-i\phi} + \frac{1}{2} \sin \theta e^{i\phi} - \frac{1}{2} \sin \theta e^{-i\phi} \right]$
 $= Ar e^{-i\omega t} \left[(1-i) \frac{1}{\sqrt{2}} Y_{11}(0, \phi) + (1+i) \frac{1}{\sqrt{2}} Y_{1-1}(0, \phi) + 2 \frac{1}{\sqrt{2}} Y_{10}(0, \phi) \right]$

(1) $\hat{L}^2 \psi = \langle \hat{L}^2 \rangle \psi$
 $= \left[\frac{1}{2} \frac{1}{\sqrt{2}} \langle 11 | + (1-i) \frac{1}{\sqrt{2}} \langle 10 | + 2 \frac{1}{\sqrt{2}} \langle 10 | \right] \hat{L}^2 \psi$
 $= \left[\frac{1}{2} \frac{1}{\sqrt{2}} \langle 11 | + (1-i) \frac{1}{\sqrt{2}} \langle 10 | + 2 \frac{1}{\sqrt{2}} \langle 10 | \right] \hat{L}^2 \psi$
 $= \frac{4\pi}{3} \cdot 2\hbar^2 + \frac{4\pi}{3} \cdot 2\hbar^2 +$

由于 ψ 由 $L=1$ 的 ψ 组成，所以 $\hat{L}^2 \psi = 2\hbar^2 \psi$

故 $\hat{L}^2 = 2\hbar^2$

(2) 由于波函数叠加原理，
 $\hat{L}^2 \psi = 2\hbar^2 \psi$

波函数叠加原理的波函数：
 $\psi(0, \phi) = \frac{1}{\sqrt{2}} Y_{11}(0, \phi) + (1-i) \frac{1}{\sqrt{2}} Y_{10}(0, \phi) + 2 \frac{1}{\sqrt{2}} Y_{10}(0, \phi)$
 $\psi(0, \phi) = \frac{1-i}{\sqrt{2}} Y_{11}(0, \phi) + \frac{1+i}{\sqrt{2}} Y_{10}(0, \phi) + 2 \frac{1}{\sqrt{2}} Y_{10}(0, \phi)$

$\hat{L}^2 = \int_0^{2\pi} \int_0^\pi \psi^*(0, \phi) \hat{L}^2 \psi(0, \phi) \sin \theta d\theta d\phi$
 $= 2\hbar^2$

$$(3) \quad P = |\langle n | \psi \rangle|^2$$

$$= \frac{1}{5}$$

$$(4) \quad P = \int_0^{2\pi} \int_0^\pi \psi^* \psi \sin\theta d\theta d\varphi$$

$$= \int_0^{2\pi} \int_0^\pi |\psi|^2 \sin\theta d\theta d\varphi$$

$$\text{IV. } \hat{H} = A\hat{\sigma}_x + B\hat{\sigma}_z$$

$$(1) \quad \text{在 } \hat{S}_z \text{ 表象中}$$

$$\hat{H} = A \cdot \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + B \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} B & -iA \\ iA & -B \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} B & -iA \\ iA & -B \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$i\frac{1}{2} \lambda' = \frac{2\lambda}{\hbar}$$

$$\begin{vmatrix} B - \lambda' & -iA \\ iA & -B - \lambda' \end{vmatrix} = 0$$

$$\lambda_1 = \frac{\hbar}{2} \sqrt{A^2 + B^2}, \quad \lambda_2 = -\frac{\hbar}{2} \sqrt{A^2 + B^2}$$

$$(2) \quad H = \frac{1}{\sqrt{2B^2 + 2B\sqrt{A^2 + B^2}}} \begin{pmatrix} iA & \\ & -iA \end{pmatrix} = \frac{1}{\sqrt{A^2 + B^2}} \begin{pmatrix} iA & \\ & -iA \end{pmatrix}$$

$$1-7 = \frac{1}{\sqrt{2B^2 + 2B\sqrt{A^2 + B^2}}} \begin{pmatrix} iA & \\ & -iA \end{pmatrix} = \frac{1}{\sqrt{A^2 + B^2}} \begin{pmatrix} iA & \\ & -iA \end{pmatrix}$$

$$(2) \quad \langle S_y \rangle = ?$$

B) 在 $\hat{S}_z$ 表象中, $\hat{S}_y$ 的矩阵元为

$$\hat{S}_y = \frac{\hbar}{4} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \lambda = \frac{\hbar}{2}, \quad 117 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \lambda = -\frac{\hbar}{2}, \quad 117 = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\text{当 } \psi = |1\rangle \text{ 时}$$

$$P_1 = |\langle 1 | 1 \rangle|^2$$

$$= \left| \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2B^2 + 2B\sqrt{A^2 + B^2}}} \begin{pmatrix} iA & \\ & -iA \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right|^2$$

$$= \left| \frac{1}{\sqrt{2}} (-i, i) \frac{1}{\sqrt{B^2 + (B-C)^2}} \begin{pmatrix} iA \\ iA \end{pmatrix} \right|^2$$

$$= \frac{(A-B+C)^2}{2[A^2 + (B-C)^2]}$$

$$\text{当 } \psi = |1\rangle \text{ 时}$$

$$P_2 = |\langle 1 | 1 \rangle|^2$$

$$= \left| \frac{1}{\sqrt{2}} (-i, -i) \frac{1}{\sqrt{A^2 + (B+C)^2}} \begin{pmatrix} iA \\ iA \end{pmatrix} \right|^2$$

$$= \frac{(A-B-C)^2}{2[A^2 + (B+C)^2]}$$

$$\text{III. } \hat{H} = \frac{72}{22}$$

$$(1) \quad \hat{L}_z \psi(\varphi) = \hbar L \psi(\varphi)$$

$$-i\hbar \frac{\partial}{\partial \varphi} \psi(\varphi) = L \psi(\varphi)$$

$$\frac{d\psi(\varphi)}{d\varphi} = \frac{iL}{\hbar} \psi$$

$$\psi(\varphi) = A e^{\frac{iL}{\hbar} \varphi}$$

$$\text{由 } \psi(\varphi) = \psi(\varphi + 2\pi) \text{ 得 } \psi(\varphi) = \psi(\varphi + 2\pi)$$

$$\psi_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}, \quad m = 0, \pm 1, \pm 2, \dots$$

$$\therefore \hat{H} = \frac{72}{22} \text{ 的矩阵元是 } \psi_m = \frac{1}{\sqrt{2\pi}} e^{im\varphi}, \quad m = 0, \pm 1, \pm 2, \dots$$

$$L_m = \frac{m^2 \hbar^2}{2I}$$

$$(2) \hat{H}' = A \sin 2\varphi$$

$$E_m = \frac{m^2 \hbar^2}{2I}, m=0, \pm 1, \pm 2, \dots, \text{则 } \psi(m) = \psi_m(\varphi)$$

① $m=0$, 为基态:

$$E_0^{(0)} = 0, E_0^{(1)} = \langle 0 | \hat{H}' | 0 \rangle = \int_0^{2\pi} \frac{1}{2\pi} A \sin 2\varphi d\varphi = 0$$

$$E_0^{(2)} = \sum_{k \neq 0} \frac{|H'_{k0}|^2}{E_0^{(0)} - E_k^{(0)}} \quad H'_{mn} = \langle m | \hat{H}' | n \rangle = \int_0^{2\pi} \frac{1}{2\pi} A \sin 2\varphi e^{i(m-n)\varphi} d\varphi$$

$$= \frac{A}{2\pi} \int_0^{2\pi} \sin 2\varphi \cdot e^{i(m-n)\varphi} d\varphi = \frac{A}{2\pi} \int_0^{2\pi} \frac{e^{2i\varphi} - e^{-2i\varphi}}{2i} e^{i(m-n)\varphi} d\varphi$$

$$= \frac{A}{4\pi i} [\int_0^{2\pi} \delta_{n, m+2} - \delta_{n, m-2}] d\varphi = \frac{A}{2i} [\delta_{n, m+2} - \delta_{n, m-2}]$$

② $m=1$, 为第一激发态:

$$\psi_0 = \psi_0^{(0)} = \frac{1}{\sqrt{2\pi}}$$

③ $m=1$, 为第一激发态:

$$\psi_1 = \psi_1^{(0)} = C_1 \alpha + C_2 \beta, \quad \alpha = \psi_1^{(0)}(\varphi), \quad \beta = \psi_{-1}^{(0)}(\varphi)$$

$$\begin{pmatrix} H'_{11} & H'_{12} \\ H'_{21} & H'_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E^{(1)} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\text{由于 } H'_{11} = H'_{22} = H'_{12} = H'_{21} = 0$$

$$\therefore H'_{11} = H'_{22} = H'_{12} = H'_{21} = 0$$

无简并态, 0 度微扰

由波函数归一化:

$$\hat{H}' | \psi^{(1)} \rangle + \hat{H}' | \psi^{(2)} \rangle = E^{(1)} | \psi^{(1)} \rangle + E^{(2)} | \psi^{(2)} \rangle + E^{(3)} | \psi^{(3)} \rangle$$

$$\therefore \langle \psi_0 | \hat{H}' | \psi^{(1)} \rangle = (E^{(1)} - E_0^{(0)}) \langle \psi_0 | \psi^{(1)} \rangle + E^{(2)} \langle \psi_0 | \psi^{(2)} \rangle$$

$$\langle \psi_0 | \psi^{(1)} \rangle = \langle \alpha | \alpha \rangle + \langle \beta | \beta \rangle$$

对于 α 和 β

$$\langle \alpha | \hat{H}' | \psi^{(1)} \rangle = E^{(1)} \langle \alpha | \psi^{(1)} \rangle$$

$$\langle \beta | \hat{H}' | \psi^{(1)} \rangle = E^{(1)} \langle \beta | \psi^{(1)} \rangle$$

$$\text{又 } | \psi^{(1)} \rangle = \sum_{k \neq 0} \frac{H'_{k0}}{E_0^{(0)} - E_k^{(0)}} | \psi_k^{(0)} \rangle$$

$$= \sum_{k \neq 0} \frac{\langle k | \hat{H}' | 0 \rangle + \langle 0 | \hat{H}' | k \rangle}{E_0^{(0)} - E_k^{(0)}} | \psi_k^{(0)} \rangle$$

$$= C_1 \sum_{k \neq 0} \frac{H'_{k0}}{E_0^{(0)} - E_k^{(0)}} | \psi_k^{(0)} \rangle$$

$$+ C_2 \sum_{k \neq 0} \frac{H'_{k0}}{E_0^{(0)} - E_k^{(0)}} | \psi_k^{(0)} \rangle$$

对于 $\psi^{(1)}$ 有 $\langle \psi_0 | \psi^{(1)} \rangle = 0$

$$\begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E^{(2)} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$G_{11} = \sum_{k \neq 0} \frac{|H'_{k0}|^2}{E_0^{(0)} - E_k^{(0)}}$$

$$= \frac{A^2}{4} \frac{1}{E_0^{(0)} - E_2^{(0)}} = -\frac{A^2}{16\hbar^2}$$

$$G_{12} = \sum_{k \neq 0} \frac{H'_{k0} H'_{k1}}{E_0^{(0)} - E_k^{(0)}} = 0$$

$$G_{21} = G_{12}^* = 0$$

$$G_{22} = \sum_{k \neq 0} \frac{|H'_{k1}|^2}{E_0^{(0)} - E_k^{(0)}} = \frac{A^2}{4} \frac{1}{E_0^{(0)} - E_2^{(0)}} = -\frac{A^2}{16\hbar^2}$$

$$= -\frac{A^2}{16\hbar^2}$$

$$\hat{H} = \frac{\omega}{2} \hat{\sigma}_z$$

$$|\psi(t)\rangle = \frac{1}{2} e^{-i\omega t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} e^{i\omega t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} \cos \omega t \\ -i \sin \omega t \end{pmatrix}$$

(1) 在 S_z 基下

归一化基矢为

$$S_1 = \frac{\hbar}{2}, |1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S_2 = -\frac{\hbar}{2}, |2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$P_1 = |\langle 1 | \psi(t) \rangle|^2$$

$$= \left| \frac{1}{\sqrt{2}} (1, 1) \begin{pmatrix} \cos \omega t \\ -i \sin \omega t \end{pmatrix} \right|^2$$

$$= \frac{1}{2} (\cos \omega t + \sin \omega t)^2$$

$$P_2 = |\langle 2 | \psi(t) \rangle|^2$$

$$= \left| \frac{1}{\sqrt{2}} (1, -1) \begin{pmatrix} \cos \omega t \\ -i \sin \omega t \end{pmatrix} \right|^2$$

$$= \frac{1}{2} (\cos \omega t - \sin \omega t)^2$$

$$\hat{H} = \frac{\omega}{2} \hat{\sigma}_z, t=0, S_z = \frac{\hbar}{2}$$

(2) $|\psi(t)\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

归一化基矢为

$$|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

S_z 的归一化基矢为

$$S_1 = \frac{\hbar}{2}, |1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S_2 = -\frac{\hbar}{2}, |2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|\psi(t)\rangle = \frac{1}{2} (|1\rangle + |2\rangle + |1\rangle + |2\rangle)$$

$$= \frac{1}{2} (|1\rangle + |2\rangle + |1\rangle + |2\rangle)$$

$$|\psi(t)\rangle = e^{-i\omega t/2} |\psi(0)\rangle$$

$$= \frac{1}{2} e^{-i\omega t/2} (|1\rangle + |2\rangle + |1\rangle + |2\rangle)$$